

The Relationship Between Study Time and Academic Performance of Students

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Abstract—This study examines the relationship between study time and academic performance among students. It aims to determine whether the number of hours spent studying each day affects students' academic achievement. Data were collected from 250 students and analyzed using Pearson correlation analysis. The results showed a correlation coefficient of -0.021 between study hours and GPA, indicating a very weak negative relationship. This suggests that study time alone does not significantly influence academic performance. Other factors such as sleep hours and social media usage were also examined. The study highlights the importance of considering multiple factors that affect student performance.

Keywords—academic performance; correlation analysis; GPA; Pearson correlation; study habits; study time

I. INTRODUCTION

Academic performance is one of the primary indicators of student success in school. Many factors may influence how well students perform academically, including study habits, attendance, motivation, sleep, and time management. Among these factors, study time is often considered an important element because it reflects the effort students devote to learning and understanding their lessons.

Many students believe that spending more time studying leads to better grades. However, academic performance may not depend solely on the number of hours spent studying. The effectiveness of study methods, concentration, and external factors can also influence learning outcomes. Because of this, it is important to examine whether there is a measurable relationship between study time and academic performance.

Previous studies have shown mixed findings regarding this relationship. Some researchers found that students who study longer tend to achieve higher grades, while others found that study quality and personal habits may be more important than the amount of time spent. This suggests that study time may not always be the strongest predictor of academic success.

This study focuses on students from National University Dasmañas and aims to determine the relationship between study hours and academic performance. Specifically, it seeks to analyze whether the number of hours students spend studying each day is significantly related to their grade point average (GPA). The findings may help students better understand how study habits affect academic achievement and may encourage more effective time management strategies.

II. REVIEW OF RELATED LITERATURE

Several studies have examined the relationship between study time and academic performance among students. Researchers have explored whether spending more hours

studying directly improves academic achievement or whether other factors contribute more significantly to student success.

A. International Literature

According to a study published by Springer, academic performance is influenced not only by the amount of time students spend studying but also by the quality of their learning strategies and self-regulated learning behaviors [1]. The study emphasized that students who effectively manage their learning process tend to perform better academically compared to students who only increase study hours without proper study methods.

Research from ResearchGate found that study time has a relationship with academic performance, but the effect is not always strong [2]. The researchers explained that excessive study time may not automatically result in higher grades if students experience stress, lack of motivation, or ineffective learning habits.

A study published in Taylor & Francis Online examined student learning behaviors and concluded that time management, class participation, and consistent study habits are important predictors of academic success [3].

Research from SAGE Journals reported that factors such as sleep quality, stress levels, and motivation significantly affect academic performance [4]. The study showed that students who maintain balanced lifestyles often perform better academically than students who focus only on increasing study time.

B. Local Literature

In the Philippine context, research on working students in the Open High School Program found that time management skills are positively associated with academic performance [6]. Students who demonstrated stronger time management capabilities tended to achieve higher grades despite their dual

responsibilities. These findings suggest that efficient use of study time, rather than the quantity of hours alone, is a key determinant of academic outcomes among Filipino students.

Additional local studies from Philippine universities highlight that student engagement, completion of academic requirements, and attendance patterns are significant contributors to GPA outcomes, reinforcing the multi-factor nature of academic performance [8].

III. METHODOLOGY

This study employs Pearson correlation analysis as its primary statistical method because the central objective is to measure the strength and direction of linear relationships among multiple continuous variables—specifically, study hours per day, sleep hours, social media usage, and academic performance measured by GPA.

Correlation is most appropriate when a researcher seeks to quantify how strongly two variables are associated without imposing a causal claim. Since this study does not manipulate any variable but instead observes naturally occurring student data, an experimental or regression-based approach would be premature. Correlation serves as a foundational analytical tool that reveals which attributes are meaningfully related.

The Pearson correlation coefficient (r) produces a value ranging from -1 to $+1$, where values near ± 1 indicate a strong linear relationship and values near 0 indicate little to no linear association. This makes it particularly well-suited for a dataset with multiple numeric attributes, as it allows for the construction of a correlation matrix that simultaneously examines all possible variable pairs across 250 student records.

Furthermore, correlation aligns with the quantitative, non-experimental nature of this research. The study does not seek to determine causality—it seeks to identify patterns and associations that can inform academic guidance and institutional decisions.

IV. RESULTS AND DISCUSSION

The results showed that the average study time of the students was approximately 5.03 hours per day, while the average GPA was 7.29. Analysis of the relationship between study time and GPA using Pearson correlation produced the following output:

```
r_value, p_value = pearsonr(data['Study_Hours_per_Day'], data['GPA'])
print('Pearson r:', r_value)
print('P-value:', p_value)

Pearson r: -0.020665961575960647
P-value: 0.7450639801780914
```

Fig. 1. Pearson Correlation Output (Study Hours vs. GPA)

The correlation coefficient of $r = -0.021$ with a p-value of 0.745 indicates a very weak negative relationship between study time and GPA. The high p-value confirms that this result

is not statistically significant, meaning the observed relationship could be due to chance. The interpretation of this result is shown below:

Interpretation: Weak Negative Correlation

Fig. 2. Interpretation of Pearson Correlation Result

Students who studied longer did not necessarily obtain higher grades. The findings indicate that academic performance may be affected by other factors beyond study time. Variables such as attendance, assignment completion, stress, sleep habits, and study methods may contribute more significantly to student outcomes.

To further understand the relationships among all variables, a full correlation matrix was computed across study hours per day, sleep hours, social media hours, and GPA. The numerical output of this matrix is presented below:

	Study_Hours_per_Day	Sleep_Hours	Social_Media_Hours	\
Study_Hours_per_Day	1.000000	-0.055822	-0.070888	
Sleep_Hours	-0.055822	1.000000	0.031571	
Social_Media_Hours	-0.070888	0.031571	1.000000	
GPA	-0.020666	0.011529	-0.042536	
				GPA
Study_Hours_per_Day	-0.020666			
Sleep_Hours	0.011529			
Social_Media_Hours	-0.042536			
GPA	1.000000			

Fig. 3. Student Performance Correlation Matrix (Numerical)

The matrix confirms that all inter-variable correlations are very weak, ranging from -0.071 to $+0.032$. None of the variables—study hours, sleep hours, or social media hours—showed a meaningful linear relationship with GPA. The heatmap visualization of this correlation matrix is presented in Fig. 4.

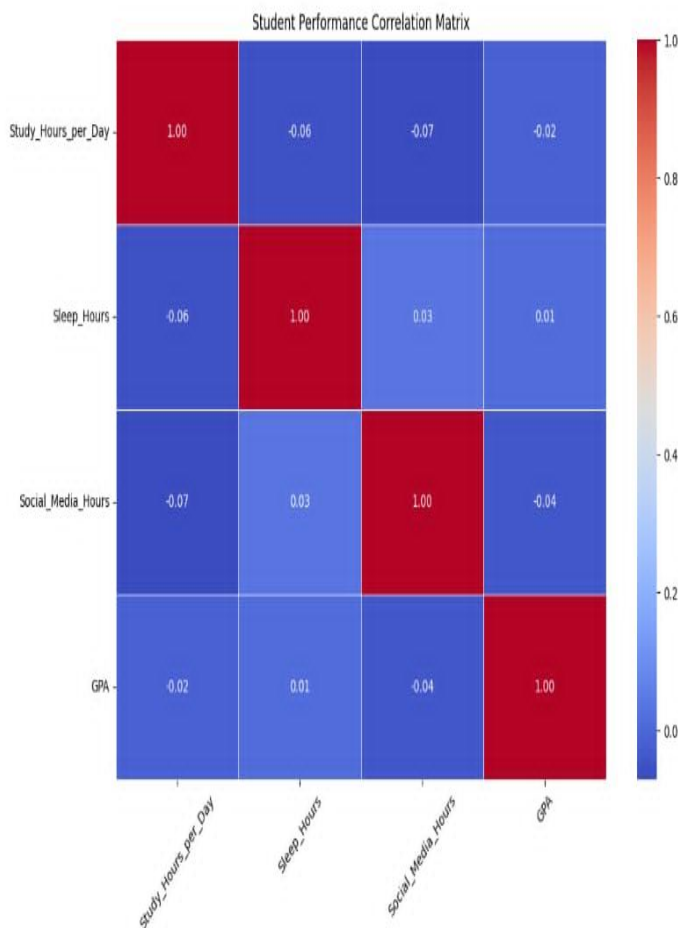


Fig. 4. Student Performance Correlation Matrix Heatmap

The heatmap uses a red-blue diverging color scale where deep red represents a perfect positive correlation ($r = 1.00$) and deep blue represents a strong negative correlation. The diagonal cells show each variable's self-correlation ($r = 1.00$). All off-diagonal cells appear in shades of blue, consistent with the weak negative correlations observed across variables. These results suggest that no single behavioral factor in the dataset strongly predicts GPA, underscoring the multidimensional nature of academic performance.

V. CONCLUSION

This study examined the relationship between study time and academic performance among students from National University Dasmañas. Based on the analysis of 250 student records, the results showed a very weak negative relationship between study hours per day and GPA, with a Pearson correlation coefficient of $r = -0.021$ ($p = 0.745$).

The findings indicate that increasing study time alone does not guarantee better academic performance. Similarly, sleep hours ($r = 0.012$) and social media usage ($r = -0.043$) showed negligible correlations with GPA. These results highlight that other factors such as learning strategies, personal discipline,

attendance, and lifestyle may play a more significant role in influencing student achievement.

The study concludes that while study time remains an important academic factor, it should be considered together with other variables to better understand student performance. Future studies may include additional factors such as motivation, study quality, and institutional support to provide a more comprehensive analysis of academic success.

REFERENCES

- [1] A. Schneider and M. Preckel, "Variables associated with achievement in higher education: A systematic review," "Self-Regulated Learning," – Springer, 2017. [Online]. Available: <https://link.springer.com/article/10.1007/s10212-021-00546-0>
- [2] M. D. Flores and R. Santos, "The relationship between students' study time and academic performance and its practical significance," – ResearchGate, 2022. [Online]. Available: <https://www.researchgate.net/publication/366691380>
- [3] P. Rivard, "How study habits influence academic outcomes," – Journal of Economic Education, Taylor & Francis Online, 2015. [Online]. Available: <https://www.tandfonline.com/doi/full/10.1080/00220485.2015.1040182>
- [4] T. Haynes, "Sleep quality, stress, and academic performance among college students," – Psychological Reports, SAGE Journals, 2006. [Online]. Available: <https://journals.sagepub.com/doi/10.2466/pr0.98.3.675-682>
- [5] [LOCAL RRL – to be added by authors]
- [6] J. A. Reyes and L. B. Cruz, "Time management skills and academic performance of working students in Open High School Program," – ResearchGate, 2022. [Online]. Available: <https://www.researchgate.net/publication/365963209>
- [7] B. Taylor and S. Nguyen, "Active learning strategies and student retention," – Internet and Higher Education, ScienceDirect, 2015. [Online]. Available: <https://www.sciencedirect.com/science/article/abs/pii/S1096751615000251>
- [8] R. Santos et al., "Factors affecting academic performance among Filipino students," – BCP Education & Psychology, 2023. [Online]. Available: <https://bcpublishing.org/index.php/EP/article/view/2696>
- [9] "Students who take breaks actually learn faster," – Times of India / Pew Research, 2025. [Online]. Available: <https://timesofindia.indiatimes.com/education/news/students-who-take-breaks-actually-learn-faster-pew-research-reveals-why/articleshow/124061411.cms>
- [10] N. J. Cepeda et al., "Distributed practice in verbal recall tasks: A review and quantitative synthesis," – Psychological Bulletin, SAGE, 2006. [Online]. Available: <https://journals.sagepub.com/doi/10.1111/j.1467-9280.2005.01641.x>
- [11] R. Struthers, J. Perry, and V. Menec, "An examination of the relationship among academic stress, coping, motivation, and performance," – Research in Higher Education, Taylor & Francis, 2000. [Online]. Available: <https://www.tandfonline.com/doi/abs/10.1080/07448480009596294>